Certified Distance Queries on Terrain Surfaces in Microseconds

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Abstract

Shortest-path distances on terrain surfaces underlie nearest-neighbour, range and route queries in geographic information systems, robotics and military planning. Exact geodesic algorithms take milliseconds to seconds per pair, and the distance oracles proposed for terrains approximate the surface by a Steiner graph, so their error is fixed by the discretisation, their indexes are large, and no answer says how wrong it can be. We present the Certified Cut Index (CCI), a distance oracle for terrain surfaces built like a hub labelling on a separator hierarchy. Every vertex stores exact geodesic distances to portals sampled on the cut lines of a kd hierarchy, and every query returns an estimate together with a certified interval $\mathrm{LB} \le d \le \mathrm{UB}$ with $(1-\varepsilon)d \le \mathrm{LB}$ and $\mathrm{UB} \le (1+\varepsilon)d$. A window lemma lets us sample the cuts instead of storing them whole, an ellipse lemma makes every label an exact geodesic computed on a small sub-mesh, and the 1-Lipschitz structure of the label rows gives the lower bound. A capped variant with certify-or-refine stores $O\big((k+\varepsilon^{-1}\log k)\, n \log n\big)$ labels and still answers every query with a certificate or exactly. On the benchmark terrains of the SE-Oracle and EAR-Oracle papers and on a 100k-vertex DEM, CCI answers vertex-to-vertex queries in 0.2–0.6 μs on one core, four orders of magnitude faster than EAR-Oracle, with mean error 0.002–0.07% against EAR-Oracle's 3.1–3.5% and an index 19–145 times smaller. Every answer we measured is certified and correct. The same index answers queries between arbitrary surface points, certified kNN and range queries, and terrains with obstacles.

1 Introduction

A terrain is usually stored as a triangulated irregular network (TIN): a triangulation of sampled points whose projection to the plane is a planar triangulation. Many queries on terrains are distance queries. A nearest-neighbour query asks for the closest objects to a location, a range query asks for all objects within a given distance, and both are routine in geographic information systems, in path planning for vehicles and robots, and in the analysis of disasters and military operations. The distance in these applications is the length of the shortest path on the surface, not the Euclidean or the projected one, which can be off by tens of percent in mountains.

Computing a single surface distance exactly is expensive. The classical algorithm of Mitchell, Mount and Papadimitriou (MMP) [18] and its practical successors [20, 21, 24] propagate a wavefront of windows over the mesh, and one query costs milliseconds on a few thousand vertices and seconds on a hundred thousand. For applications that ask many queries, the database community has proposed distance oracles for terrains: SE-Oracle [23], EAR-Oracle [11], RC-Oracle [25] and UP-Oracle [26]. They replace the surface by a Steiner graph and index that graph with well-separated pair decompositions or spanners. Their guarantee is relative to the Steiner graph, and we measured mean errors of 0.8–3.5% and maximum errors up to 25% against exact geodesics, with indexes of hundreds of megabytes already at 10k vertices. More importantly, a returned distance comes with no indication of its error.

Road networks faced the same problem and solved it with labelling indexes. H2H [19] takes a tree decomposition of the graph and lets every vertex store its distances to the separator vertices of its ancestors, so that a query is a minimum over one separator. We ask whether the same architecture works on a continuous surface, and what it takes to keep it exact or certified. Two things change. A separator of a surface is a curve, not a finite set of vertices, so the index can only store distances to finitely many sampled points of it. And labels must be geodesic distances on the surface, which are expensive to compute for every vertex and portal pair.

We propose the Certified Cut Index (CCI). Its separators are axis-parallel cut lines of a kd hierarchy [6] over the chart of the terrain. Every vertex stores exact geodesic distances to portals sampled on the cuts of its ancestors, at an arc gap g near the region and at geometrically growing gaps away from it. For a query, the minimum over one cut gives an upper bound that is the length of an actual path, and the 1-Lipschitz structure of the label rows along the cut gives a lower bound. Both are within a factor $1 \pm \varepsilon$ of the true distance, so every answer carries a certificate. Our contributions are as follows: (i) the Certified Cut Index and its $(1+\varepsilon)$ guarantee, built on a window lemma that bounds where a shortest path can cross a cut, an ellipse lemma that makes labels computable exactly on small sub-meshes, and an interval lower bound read from the labels (Section 4); (ii) a capped variant with certify-or-refine whose labels are near-linear in n, and certified queries between arbitrary surface points (Sections 4.4 and 4.5); (iii) engineering that keeps the bounds sound while making the index fast and compact: hybrid exact sweeps, closed-loop 8/16-bit label coding, a parallel build with bit-identical labels, and certified kNN and range queries (Section 5); (iv) an extension to weighted terrains and terrains with obstacles; and (v) an evaluation against every terrain oracle with available code, run with the authors' code on the same machine and the same queries (Section 6).

2 Related work

Exact and approximate geodesics. MMP [18] computes single-source geodesics exactly by propagating windows of constant unfolding over the mesh; Chen and Han [7] gave an $O(n^2)$ algorithm, and the practical implementations of Surazhsky et al. [21], ICH [24] and the triangle-oriented VTP [20] made exact geodesics routine. We use the first and the last to compute labels. Fast marching [15] and the heat method [8] approximate distances by solving PDEs on the mesh. The saddle vertex graph [27] and its successor DGG [2] precompute a sparse graph whose shortest paths approximate geodesics very well, but a query still runs Dijkstra on a graph with more edges than the mesh, and takes milliseconds.

Terrain distance oracles in databases. SE-Oracle [23] places Steiner points on the edges and builds a compressed partition tree with well-separated node pairs, answering POI-to-POI queries with a

$(1+\varepsilon)$ guarantee relative to the Steiner graph. EAR-Oracle [11] extends this to arbitrary points with highway nodes on an adaptive grid, and adds an additive error for points between Steiner points. RC-Oracle [25] targets point clouds and UP-Oracle [26] terrains that change, both for a set of POIs. All of them measure their distance on a discretised surface. Kaul et al. [12] derived lower and upper bounds for terrain distances with a fixed ratio of about 1/0.9, computed per pair by a graph search, and used them to prune kNN candidates; K-Algo [13] computes exact paths by a search that refines a Steiner graph on the fly. Our bounds are read from an index in a microsecond and their ratio is controlled by ε.

Oracles on polyhedral surfaces with theory. Aleksandrov et al. [4, 5] and Djidjev and Sommer [9] built $(1+\varepsilon)$ oracles for weighted polyhedral surfaces from a Steiner discretisation with geometrically placed points and shortest-path separators of the resulting graph. Djidjev and Sommer achieve near-linear space, $O(n\varepsilon^{-3/2}\log^2\frac{n}{\varepsilon}\log^2\frac{1}{\varepsilon})$, and query time $O(\varepsilon^{-1}\log\frac{1}{\varepsilon}+\log\log n)$. Near-linear space is therefore known in theory, and our uncapped index is asymptotically larger. As far as we know, neither line of work has an implementation, their constants include the Steiner density and the smallest face angle, and neither returns a per-query interval.

Distance labels and portals on graphs. Planar graphs have small separators [17], and the approximate oracles of Klein [16], Thorup [22] and Kawarabayashi et al. [14] store distances to portals on shortest-path separators; Thorup's ε-covers keep $O(1/\varepsilon)$ portals per vertex and separator. Exact distance labels on planar graphs need $\Theta(\sqrt{n})$ entries per vertex in the worst case [10], which is the graph analogue of our $n^{1.5}$ total. On road networks, hub labels [1], pruned landmark labelling [3] and H2H [19] answer exact queries in microseconds. CCI transfers H2H's architecture to a continuous surface: cut lines instead of vertex separators, sampled portals instead of whole bags, exact geodesic labels, and a certified interval instead of an exact minimum.

Learned distances. NeuroGF [28] fits a neural field to one mesh and answers a query with one forward pass. It has no error bound, and its fixed capacity has to cover the whole surface.

3 Preliminaries

A terrain T is a triangulated height field: a triangulation of points (x_v, y_v, z_v) whose projection to (x, y) is a planar triangulation of a domain $U \subset \mathbb{R}^2$, the chart. We write n for the number of vertices. Each face f carries a cost $w_f \geq w_{\min} > 0$; an unweighted terrain has $w \equiv 1$. The cost $L(\gamma)$ of a curve γ on T is $\sum_f w_f$ times its 3-D length inside f, and $d(p, q)$ is the infimum over curves from p to q. Shortest paths exist by compactness. For a point p of T we write p also for its chart position and P for the point in space.

Two lower bounds are used throughout. Put $\lambda := w_{\min}$. On a face, the metric pulled back to the chart is $w_f^2(I + \nabla z \nabla z^\top) \succeq w_f^2 I$, and 3-D length dominates the chord, so for every curve γ from p to q

$$L(\gamma) \;\geq\; \lambda\, L_{\text{chart}}(\gamma) \;\geq\; \lambda\,|p-q|, \qquad L(\gamma) \;\geq\; \lambda\,|P-Q|. \quad (1)$$

The index has two parameters: the accuracy $\varepsilon \in (0, 1)$ and a portal gap $g > 0$, an arc length on the surface. They define the near radius $R := g/(\varepsilon\lambda)$. We consider vertex-to-vertex (V2V) queries and arbitrary-point (A2A) queries, and measure the relative error of an estimate $\hat{d}$ as $|\hat{d}/d - 1|$.

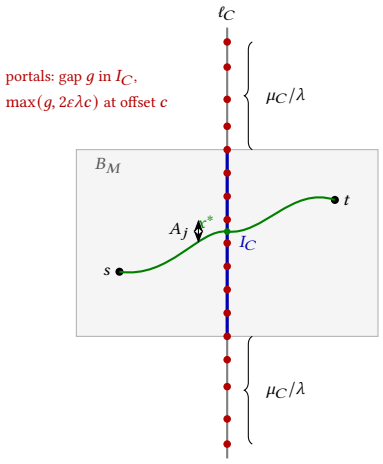


Figure 1: One cut of the index. The region M has bounding box B_M, and the cut line ℓ_C meets it in I_C. A shortest path from s to t must cross ℓ_C within the window, at most μ_C/λ from I_C (Lemma 4.1). Portals are placed at arc gap g inside the box and at gaps growing with the offset outside it. Both s and t store exact geodesic distances to all portals.

4 The Certified Cut Index

4.1 Cut hierarchy and windows

The index is built on a kd hierarchy of the vertices. A region M is a set of vertices with chart bounding box B_M. If the diagonal of B_M is at most R, the region is a leaf. Otherwise we cut it along the axis a of the larger extent at a value v strictly between the two middle member coordinates, chosen so that no vertex of T has a-coordinate v. The cut line is $\ell_C = \{x_a = v\}$, and the children are the members on its two sides. We write $I_C = \ell_C \cap B_M$.

A shortest path between two members of M on opposite sides of the cut must cross ℓ_C (by continuity of its chart projection), but it may leave the box first. The window bounds how far. Pick a hub $h \in M$ near the centre of the box and let $\mu_C = \max_{u\in M} \tilde{U}(h, u)$, where $\tilde{U}$ is the mesh-graph distance, an upper bound of d because a graph path is a curve on T.

Lemma 4.1 (window). *Let $s, t \in M$ lie on opposite sides of ℓ_C, let γ be a shortest path between them and $x^* \in \gamma \cap \ell_C$. Then $d(s,t) \geq 2\lambda\ \mathrm{dist}(x^*, I_C)$, and $\mathrm{dist}(x^*, I_C) \leq \mu_C/\lambda$.*

Proof. By (1), $d(s,t) \geq \lambda(|s-x^*| + |x^*-t|) \geq 2\lambda\,\mathrm{dist}(x^*, B_M)$, and for a point of ℓ_C the nearest point of B_M lies on I_C. On the other side, $d(s,t) \leq d(s,h) + d(h,t) \leq 2\mu_C$. □

The window $W_C = \{x \in \ell_C : \mathrm{dist}(x, I_C) \leq \mu_C/\lambda\}$ therefore contains every crossing that matters. Its intersection with the surface is a union of segments, the runs, which we compute exactly from the faces the line crosses. On each run the lifted cut is a polyline, and we measure arc length $A(\cdot,\cdot)$ along it.

Algorithm 1 Query (s, t)

1: **if** $|s - t| \le R$ **then return** $d(s, t)$ from the near table of s
2: **end if**
3: $C \leftarrow$ the deepest cut whose region contains s and t
4: $f_j \leftarrow \ell(s, p_j) + \ell(t, p_j)$ for every portal $p_j \in P_C$
5: UB $\leftarrow \min_j f_j$
6: LB $\leftarrow \max\left(\lambda|S - T|,\ \min_j[\frac{1}{2}(f_j + f_{j+1}) - A_j]\right)$ ▷ consecutive portals of a run
7: $\hat{d} \leftarrow$ the interpolated minimum of f, clamped into [LB, UB]
8: **return** $(\hat{d}, \text{LB}, \text{UB})$

4.2 Portals and labels

The portals P_C of a cut are points of the runs, sorted along the line, including the ends of every run and of I_C. For two consecutive portals of a run with arc distance A_j we require

$$A_j \le g \text{ if the interval lies in } I_C, \qquad A_j \le \max(g,\ 2\varepsilon\lambda c_j) \text{ otherwise,} \tag{2}$$

where c_j is the chart distance from the interval to I_C. Inside the box the portals are dense; outside it the gap grows linearly with the offset, which Lemma 4.1 allows because a path crossing at offset c is at least $2\lambda c$ long. The tails therefore hold only $O(\varepsilon^{-1}\log(\mu_C/\lambda R))$ portals.

Every member u of M stores the exact distances $\ell(u, p) = d(u, p)$ to all portals $p \in P_C$, and the rows of all cuts on the path from u's leaf to the root form its label. In addition, every vertex stores a near table with the exact distances to all vertices within chart distance R.

Exact labels on small sub-meshes. A label is a geodesic distance on the whole surface, but it can be computed on a small part of it.

Lemma 4.2 (ellipse). *If $d(x, u) \le D$, every point of a shortest path from x to u lies within chart distance $D/(2\lambda)$ of the segment $[x, u]$.*

Proof. By (1), a point z of the path has $|z-x| + |z-u| \le D/\lambda$, so it lies in the ellipse with foci x, u and major axis D/λ, whose points are within its semi-minor axis $\frac{1}{2}\sqrt{(D/\lambda)^2 - |x-u|^2} \le D/(2\lambda)$ of the segment. □

With D a graph upper bound through the hub, all label geodesics of a cut stay inside the bounding box of $M \cup P_C$ dilated by $D/(2\lambda)$. We take the faces that meet this box, insert the portals as vertices (on their faces' planes, so the surface does not change), and run one exact single-source sweep per portal on this sub-mesh. A geodesic of T inside the sub-mesh is a curve of the sub-mesh and conversely, so the sub-mesh distances are exact. The near tables are computed the same way, from one sweep per vertex on its own small sub-mesh.

4.3 Query and certificate

Algorithm 1 answers a query. If s and t are not near, they lie in different leaves, hence on opposite sides of the deepest cut C whose region contains both. Every f_j is the length of a path through p_j, so UB $= \min_j f_j \ge d(s, t)$. The lower bound uses that a label row is 1-Lipschitz along the cut in arc length: $|\ell(u, p) - \ell(u, q)| \le A(p, q)$.

Lemma 4.3 (sandwich). LB $\le d(s, t) \le$ UB *for every pair.*

Proof. Near pairs are exact. Otherwise let x^* be a crossing of a shortest path, which lies on a run between consecutive portals p_j, p_{j+1} by Lemma 4.1, and let $a = A(p_j, x^*)$. The cut itself is a curve, so $d(x^*, p_j) \le a$, and $f_j \le d(s, x^*) + a + a + d(x^*, t) = d(s, t) + 2a$. Likewise $f_{j+1} \le d(s, t) + 2(A_j - a)$. Adding gives $\frac{1}{2}(f_j + f_{j+1}) - A_j \le d(s, t)$, and the chord term is a lower bound by (1). □

Theorem 4.4. *For every pair of vertices, $(1 - \varepsilon)\, d(s, t) \le$ LB $\le d(s, t) \le$ UB $\le (1 + \varepsilon)\, d(s, t)$, with equality when $|s - t| \le R$.*

Proof sketch. For a non-near pair, $d := d(s, t) \ge \lambda|s - t| > \lambda R = g/\varepsilon$ by (1). From the proof of Lemma 4.3, UB $\le \min(f_j, f_{j+1}) \le d + A_j$. If the crossing interval lies in I_C, $A_j \le g < \varepsilon d$. Otherwise its offset satisfies $d \ge 2\lambda c_j$ by Lemma 4.1, and (2) gives $A_j \le \max(g, 2\varepsilon\lambda c_j) \le \varepsilon d$. The lower bound follows the same way: an interval term is at least $d - A_j$, and a tail interval with $A_j > \varepsilon d$ has both portals at offset more than $d/(2\lambda)$, where every f is at least $2\lambda c_j > d$. The full proof is in Appendix A. □

The certificate is not only a worst-case statement. UB/LB is known for every answer, and it is usually far below $1 + \varepsilon$: the median interval width is 0.7–2% for vertex pairs in our experiments. The returned estimate is the interpolated minimum of the label sum, which in our experiments is 3–4 orders of magnitude more accurate than ε.

4.4 Capped portals and certify-or-refine

The top cuts are long and have $\Theta(\sqrt{n})$ portals each, and every vertex stores them, so the labels grow like $n^{1.5}$ in total. The capped variant puts at most k portals inside the box of each cut: its gap is $g_C = \max(g, A(I_C)/k)$, and its tails step by $\max(g_C, 2\varepsilon\lambda c)$. Lemma 4.3 does not use the gap, so the interval stays sound. Theorem 4.4 needs $g_C \le \varepsilon d$, which fails only for pairs closer than g_C/ε across a capped cut. For these the query computes the distance exactly, with one sweep on the faces meeting bbox(s, t) dilated by UB$/(2\lambda)$ (Lemma 4.2).

Theorem 4.5. *In the capped index every answer satisfies* UB $\le (1 + \varepsilon)$LB *or is exact. A cut has at most $k + O(\varepsilon^{-1}\log k)$ portals, so the labels hold $O\big((k + \varepsilon^{-1}\log k)\, n \log n\big)$ entries.*

The tail count follows from $g_C \approx \sigma A(I_C)/k$: a tail first steps by g_C for $O(1/\varepsilon)$ portals and then grows geometrically up to the window, whose length is about $A(I_C)/\lambda$, which takes $O(\varepsilon^{-1}\log k)$ portals. A uniformly random pair needs the exact route with probability about $2\pi/(k\varepsilon)^3$, and we measured no refined pair at all among 1,500 pairs at 10k vertices and 200 at 100k.

4.5 Arbitrary surface points

For a point p on face f with corners a_i and a point q on face f' with corners b_j, the straight segment inside a face is a path, so $d(p, a_i) \le w_f|P - A_i|$. The vertex certificates and the triangle inequality give

$$\text{UB}(p, q) = \min_{i,j}\ w_f|P - A_i| + \text{UB}(a_i, b_j) + w_{f'}|Q - B_j|,$$

$$\text{LB}(p, q) = \max\left(\lambda|P - Q|,\ \max_{i,j}\ \text{LB}(a_i, b_j) - w_f|P - A_i| - w_{f'}|Q - B_j|\right).$$

Pairs within R, and pairs whose interval is wider than $1 + 2\varepsilon$, are computed exactly by one sweep between the two points on an ellipse sub-mesh.

Table 1: Size, query and build of the two variants. D: side of the chart; $\bar{\sigma}$: mean metric speed along the cuts; ρ: vertex density; K_C: portals of the answering cut. Build times assume a sweep over m faces costs $O(m \log m)$.

	kd (Theorem 4.4)	capped (Theorem 4.5)
labels	$\Theta(n^{1.5})$; $\approx 3\bar{\sigma}D/g$ per vertex	$O((k+\varepsilon^{-1}\log k)\, n \log n)$
near tables	$\pi\rho g^2/(\varepsilon\lambda)^2$ per vertex	same
query	$O(\text{depth} + K_C)$	same, plus a rare exact sweep
build	$\Theta(n^{1.5}\log n)$	$O((k+\varepsilon^{-1}\log k)\, n \log^2 n)$

Theorem 4.6. *Every A2A answer satisfies* $\text{LB} \le d(p,q) \le \text{UB} \le (1+2\varepsilon)\text{LB}$.

4.6 Complexity

Table 1 summarises the costs. A cut at level ℓ of the hierarchy sweeps sub-meshes of $\Theta(n/2^\ell)$ faces, one per portal, and there are $O(\log n)$ levels. Uncapped, the top cuts dominate: $\Theta(\sqrt{n})$ portals each over $\Theta(n)$ faces. The cap bounds the number of sweeps per cut by the number of portals, so the capped build is near-linear as well. The near tables add $O(n\varepsilon^{-2}\log n)$ to both. We state plainly that the asymptotic space of Djidjev and Sommer [9] is smaller than that of our uncapped index; our contribution is an implemented index that certifies every answer, with constants we measure.

5 Engineering

Hybrid exact sweeps. We use two exact solvers: Kirsanov's implementation of MMP [21] and VTP [20]. VTP is 2.6–4.1 times faster per full sweep, but its vertex queue does not make a popped vertex final, so it has no exact early stop. MMP processes windows in order of their minimum distance, and once that minimum exceeds the current value at every target no later window can improve any of them, so it stops early without changing a single distance. For each cut we choose MMP when the sub-mesh is large relative to the members' box, and VTP otherwise. VTP needs its per-sweep state reset and consistently oriented faces; it links its window lists from the first edge, so it also rejects sub-meshes that fall into several pieces, which we sweep piece by piece. On Gunnison10k the hybrid build takes 202 s, against 284 s with MMP alone and 215 s with VTP alone, with identical labels.

Closed-loop label coding. Labels are stored as float32 by default, rounded up so that they stay upper bounds bit-exactly, and lower bounds subtract the matching relative margin. Most of the index is label rows, and neighbouring entries of a row differ by at most the arc gap between their portals (the rows are 1-Lipschitz). We therefore code a row in runs of deltas with 16 or 8 bits, anchored by a float32 value per run, and encode in a closed loop: each code is chosen from the decoded value of the previous entry, so the error does not accumulate. Every decoded value is at least the stored one, so the upper bound stays an upper bound; the lower bound subtracts the bounded decoding error, and the estimate is de-biased by half of it. A block scan over 8 portals at a time with four accumulators keeps the decoding cost low. The near tables are stored once per pair, in the row of the smaller vertex id, with varint id deltas; ratio-coded variants with 16 and 8 bits store $D = |PQ|(1 + c\,q)$ with the least code c that is not below the true distance.

Parallel build. The top cuts spread their portals over the threads and the remaining cuts go one per thread. The labels are bit-identical to a single-threaded build, two threads give a 2.0× speed-up, and the peak memory is about the coded index size because rows are coded as soon as a cut is done. A memory gate bounds the solvers alive at once. All timings in this paper use one thread.

Certified proximity queries. For a kNN or range query from s over a set of objects, the index gives an interval for each candidate. An object whose UB is below the k-th smallest LB is in the answer, one whose LB exceeds the k-th smallest UB is not, and only the ambiguous ones need an exact sweep from s, which we run once for all of them on an ellipse sub-mesh. Without that sweep, returning the k objects of smallest UB is a certified ε-approximate kNN: every returned object has $d \le (1+\varepsilon)d_k$, with d_k the true k-th distance. Likewise, the objects with UB $\le r$ form a range answer with no false positives that contains every object within $r/(1+\varepsilon)$.

Weighted terrains. Exact weighted geodesics have no practical algorithm, so on weighted terrains the labels are distances in a Steiner graph with m points per edge, shared by all searches and searched with a stop once all members of a cut are settled. Graph paths are curves on the surface, so UB is still the cost of an actual path, but the labels are not exact, so the interval lower bound no longer holds and we keep only the chord bound $\lambda|S - T|$. The weighted index therefore returns an upper bound and an estimate, not a $(1+\varepsilon)$ certificate.

Obstacles. An obstacle removes the faces whose chart centroid lies inside it, so the surface gets a hole and paths go around it. The certificate needs no change. A hole only lengthens paths, so every chart lower bound still holds; μ_C is a graph distance on the holed mesh, so a detour widens the window by itself; a cut line broken by holes gives several runs, each bracketed by portals at its ends; and the near tables hold every pair within chart radius R with its exact distance, including pairs close in the chart but separated by a wall. One practical point matters. Where two holes touch at a single vertex the mesh is no longer a manifold, and there Kirsanov's MMP returned distances up to 5% too long, longer than a path in a Steiner graph, while VTP stayed exact. We therefore separate touching holes by keeping the largest fan of faces at such vertices; on the cleaned meshes MMP and VTP agree to 10^{-15}.

6 Experiments

Setup. The experiments of this version run on an Apple M1 laptop (8 GB), on one thread. Our index is implemented in C++17. The ground truth is exact MMP in double precision. We use the four benchmark terrains of the SE-Oracle and EAR-Oracle papers (HorseMount, BigMount, HeadlightMount, RobinsonMountain, 793–3,696 vertices), Gunnison10k (9,610 vertices) and a 100k-vertex DEM, GunnisonForest100k. Our index and DGG answer the same 1,500 random vertex pairs per terrain (seed 7). EAR-Oracle and the other methods of its authors' harness answer 100 pairs with the points moved 5 cm into a face, because the harness cannot locate a point

exactly at a vertex. The portal gap g is given in mean edge lengths, and we report ε and g per experiment. A larger evaluation on terrains with 0.5–2.5M faces is in progress (Section 6.8).

Competitors. Every competitor runs with its authors' code: SE-Oracle, EAR-Oracle, K-Algo and the on-the-fly Steiner schemes in the harness of the EAR-Oracle authors [11], with a fix for weighted endpoints that changes their published weighted errors; UP-Oracle, WSPD-Oracle and RC-Oracle from the UP-Oracle code [26]; DGG [2] with a port to our platform; NeuroGF [28] with the authors' network and training, trained on exact distances and paths of our terrains; and exact MMP and VTP on the fly.

6.1 Main comparison

Table 2 and Figure 2 compare the methods.

Query time. CCI answers in 0.18–0.36 μs. EAR-Oracle takes 1.3–10.2 ms, so we are 7,000–28,000 times faster, and DGG, which runs Dijkstra on its graph, takes 1.4–22 ms. An exact VTP sweep per query costs 3.3–55 ms, and K-Algo and the on-the-fly Steiner schemes are slower still (Figure 2).

Accuracy. With $\varepsilon = 0.1$ our mean error is 0.0076–0.024% and the maximum 0.37–1.79%, while EAR-Oracle has a mean error of 3.1–3.5% and a maximum of 6.4–24.5%. Its nominal ε is relative to its Steiner graph, and the Steiner graph with 5 points per edge is itself about 3% longer than the surface. Increasing the number of Steiner points to 10 and 20 on HorseMount lowers EAR-Oracle's mean error only to 2.5% and 2.3%, at 541 and 683 MB. SE-Oracle reaches 0.81% after 1.7 h of build and 942 MB on HorseMount. DGG is the most accurate approximate method, with mean errors of about 0.0025%, but its queries are four orders of magnitude slower than ours and return no bound.

Certificates. All 7,500 answers in Table 2 are certified and correct. The guaranteed bound is never violated, and it is about two times conservative in the worst case: with $\varepsilon = 0.1$ the largest UB error over all pairs is 0.9–3.4%, against the guaranteed 10%.

Size and build. The index is 1.3–37 MB, 19–145 times smaller than EAR-Oracle's, and builds 3–8 times faster. DGG's graph is two to three times larger than our index and builds faster.

6.2 Size against error

Figure 3 shows the size against the mean error over a sweep of ε and g. Our fronts lie entirely below and to the left of EAR-Oracle: on Gunnison10k the coarsest setting, $\varepsilon = 1$ at 21 MB, has a mean error of 0.31%, ten times lower than EAR-Oracle's best, at a 30th of its size. Pushing ε up to 3.2 with a fixed near radius brings our error to EAR-Oracle's level: 3.7% at 0.66 MB on HorseMount against EAR-Oracle's 3.1% at 189 MB, and 2.4% at 3.9 MB on RobinsonMountain against 3.5% at 672 MB. Every answer stays certified. On Gunnison10k, $\varepsilon = 0.05, 0.1, 0.2$ give 64, 37 and 24 MB with mean errors of 0.0024%, 0.0076% and 0.023%.

100k vertices. Table 3 gives the storage variants on Gunnison-Forest100k with $\varepsilon = 0.2$. The float32 index takes 534 MB, the 8-bit codes reduce it to 224 MB with the mean error going from 0.0069% to 0.0075%, and a cap of 50 portals per cut brings it to 177 MB at 0.041%. No pair needed refining. EAR-Oracle did not finish on this terrain: it ran for 14.3 h on the 8 GB machine without producing an index. ε_c-covers in the style of Thorup [22], which keep for every vertex only the portals needed to reach the rest of the cut, are dominated by the 8-bit codes at every size we tried: at 100k, covers with $\varepsilon_c = 0.05$ take 230 MB for a mean error of 0.79%. The shifted hierarchy, three shifted quadtrees with a gap that grows with the cell size, is 1.4 times larger than the kd hierarchy at 100k vertices, as its constants predict; it can pay off only at larger sizes.

Growth with n. From 0.8k to 100k vertices the build time grows like $n^{1.45}$ and the index like $n^{1.32}$, close to the $n^{1.5}$ of Table 1; the 100k build takes 41–66 min on the laptop, depending on the load. The capped variant and the larger terrains are the subject of Section 6.8.

6.3 Arbitrary points

Table 4 gives the certified A2A queries. Every answer is sound, the mean error is 0.03–0.14%, and a query takes 78–123 μs, because the four corner pairs are looked up and a few pairs are refined. The refine step bounds the worst case: on BigMount one pair in 100 was refined, and the largest UB error fell from 20.2% to 10.2%. On the same kind of queries EAR-Oracle has a mean error of 4.9% and a maximum of 87% on HorseMount, at 4.4 ms per query.

6.4 Weighted terrains

On HorseMount with the face weights of the EAR-Oracle benchmark, our index with a Steiner graph of 5 points per edge has a mean error of 0.35%, a build time of 10.9 s, a size of 2.9 MB and a query time of 1.3 μs, measured against the shortest path found by any method. EAR-Oracle with 5 Steiner points has 2.42% at 25.9 s, 166 MB and 1.14 ms. Against a Steiner graph with 24 points per edge as the reference, our C++ index has a mean error of 0.38% with 5 points per edge and 0.23% with 10.

6.5 Obstacles

Table 5 gives a pilot on Gunnison10k with the obstacles of Figure 4. Every answer is certified and correct, uncapped and capped, over random, near, distance-band and A2A pairs, including the 208 pairs whose shortest path on the walls mesh is at least 20% longer than the straight line. The mean error is 0.03–0.04%, as without obstacles. On the walls mesh EAR-Oracle has a mean error of 3.59% and a maximum of 7.26%, with 36 ms per query and an index of 173 MB, and DGG ($\varepsilon = 10^{-3}$) has a mean error of 0.027% at 8.4 ms per query.

6.6 Proximity queries

On HorseMount, a certified kNN query with $k = 50$ over all vertices takes 27 μs in the approximate mode, against 0.5 ms for the exact answer from the index with refine and 0.85 ms for an on-the-fly search pruned by chart distances. On Gunnison10k with random objects the approximate mode answers kNN in 5 μs for $k = 1$, 20 μs for $k = 10$ and 165 μs for $k = 100$, and range queries in 2–7 μs. In all our runs its recall against the exact answer was at least 99.2%, and the guarantees of Section 5 held on every query. The exact mode needs one sweep only for the objects near the boundary of the answer; at 100k vertices it is 60 times faster than the pruned

Table 2: Vertex-to-vertex queries, one thread. Ours: $\varepsilon = 0.1$, $g = 0.5$ edges, hybrid sweeps, float32 labels. Columns: build time / index size / query time / mean error / maximum error. Every answer of ours is certified and correct (LB $\le d \le$ UB on all pairs); none of the other methods returns a bound.

terrain	n	ours	EAR-Oracle (sp 5)	DGG ($\varepsilon = 10^{-4}$)	VTP
HorseMount	793	3.1 s / 1.3 MB / 0.18 μs / 0.024% / 1.16%	23.5 s / 189 MB / 1.3 ms / 3.10% / 24.5%	0.73 s / 4.4 MB / 1.4 ms / 0.0023% / 0.02%	3.3 ms
BigMount	1,440	10.4 s / 3.3 MB / 0.23 μs / 0.022% / 0.86%	52.4 s / 319 MB / 1.8 ms / 3.19% / 6.41%	1.5 s / 8.9 MB / 2.6 ms / 0.0025% / 0.02%	6.6 ms
HeadlightMount	2,470	29.1 s / 6.5 MB / 0.25 μs / 0.016% / 1.30%	107 s / 592 MB / 2.5 ms / 3.29% / 8.47%	2.9 s / 16.8 MB / 5.5 ms / 0.0026% / 0.02%	12.0 ms
RobinsonMountain	3,696	44.1 s / 10.7 MB / 0.28 μs / 0.020% / 1.79%	188 s / 672 MB / 3.8 ms / 3.50% / 8.60%	4.5 s / 25.3 MB / 7.9 ms / 0.0023% / 0.02%	18.1 ms
Gunnison10k	9,610	202 s / 37.4 MB / 0.36 μs / 0.0076% / 0.37%	672 s / 702 MB / 10.2 ms / 3.43% / 9.37%	13.2 s / 72.8 MB / 22.3 ms / 0.0028% / 0.01%	55.0 ms

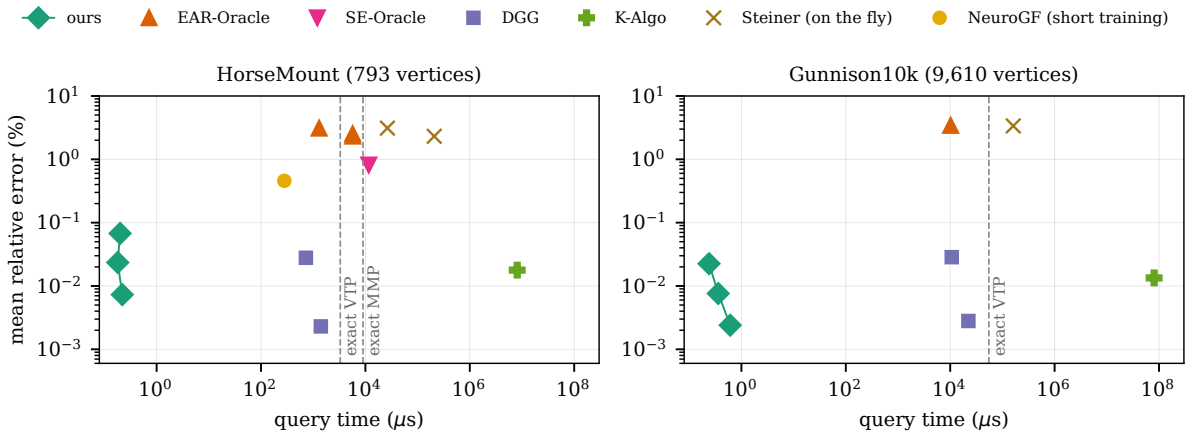


Figure 2: Query time against mean error for every method we ran, on the smallest benchmark terrain and on Gunnison10k. The three points of ours are $\varepsilon = 0.05, 0.1, 0.2$. Dashed lines mark the exact solvers, which have no error. NeuroGF was trained with a short budget on HorseMount only; its full-budget runs are in progress.

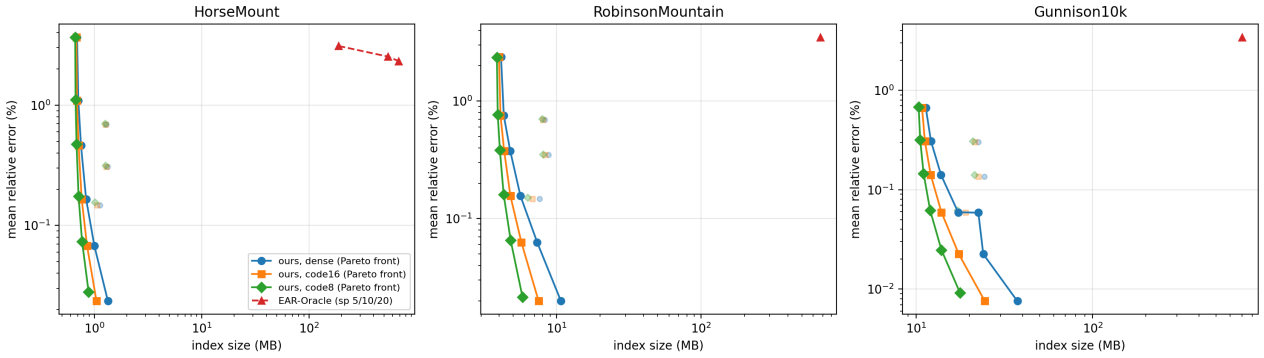


Figure 3: Index size against mean error. Each curve is the Pareto front of our index over ε and g for one label storage (float32, 16-bit and 8-bit codes); faded points are dominated settings. Triangles are EAR-Oracle with 5, 10 and 20 Steiner points per edge.

on-the-fly search for $k = 1$ and 1–2 times faster for large k, where that sweep dominates.

6.7 The POI setting

UP-Oracle and its baselines answer queries between a fixed set of points of interest. In that setting the simplest exact index is a $P \times P$ table built by one VTP sweep per POI. It is exact, sub-microsecond, and the cheapest to build for up to about 1,000 POIs on our terrains, which neither UP-Oracle nor our index beats. Our index in POI mode, which indexes only the POIs, becomes smaller than the table from about 1,000 POIs, and at 3,000 POIs on Gunnison10k it takes 3.1 MB against the table's 36 MB. Our index is designed for queries between all vertices and arbitrary points, where no table fits.

Table 3: Storage variants on GunnisonForest100k (100,511 vertices), $\varepsilon = 0.2$, one thread.

hierarchy	labels	size	mean error	query
kd	float32	534 MB	0.0069%	0.38 μs
kd	16-bit	330 MB	0.0069%	–
kd	8-bit	224 MB	0.0075%	1.3 μs
cap 100	8-bit	198 MB	0.017%	–
cap 50	float32	346 MB	0.039%	–
cap 50	8-bit	177 MB	0.041%	0.64 μs
kd	covers $\varepsilon_c = 0.05$	230 MB	0.79%	–

Table 4: Certified A2A queries, $\varepsilon = 0.1$, 100 random pairs of surface points per terrain.

terrain	query	mean err.	max err.	max UB err.
HorseMount	123 μs	0.11%	0.63%	6.2%
BigMount	111 μs	0.14%	2.69%	20.2% (10.2%)
HeadlightMount	89 μs	0.076%	0.77%	5.9%
RobinsonMountain	85 μs	0.11%	0.97%	11.4%
Gunnison10k	78 μs	0.034%	0.23%	3.3%

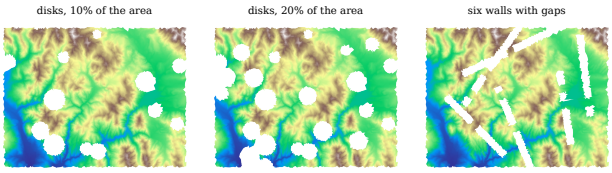


Figure 4: The obstacle meshes of the pilot on Gunnison10k. The faces inside the obstacles are removed.

Table 5: Obstacles on Gunnison10k (Figure 4), ours with $\varepsilon = 0.2$, $g = 1$ edge, cap 64. The last column counts the random pairs whose distance is at least 1.2 times their straight-line distance.

obstacles	build	mean err.	max UB err.	sound	pairs ≥ 1.2
disks, 10%	61 s	0.033%	4.3%	yes	5
disks, 20%	55 s	0.044%	1.1%	yes	20
six walls	131 s	0.032%	2.2%	yes	208

6.8 Large-scale evaluation (in progress)

We are running the full evaluation on the UP-Oracle terrains (six terrains at 0.5M, 1M, 1.5M, 2M and 2.5M faces) on an HPC cluster with Intel Xeon Gold 6248R processors, again with one thread per method. It covers

- the six terrains at 0.5M faces, uncapped, with cap 64 and with $\varepsilon = 0.1$; the cap sweep $k = 8, \ldots, 256$; the size-error curve against EAR-Oracle; the solver and storage ablations;
- scaling from 0.5M to 2.5M faces, uncapped, capped and with the shifted hierarchy, and the thread scaling of the parallel build;
- weighted terrains, A2A queries, distance bands, kNN and range queries, and the POI setting of UP-Oracle;
- the competitors at scale: EAR-Oracle, DGG, NeuroGF with its full training budget, the UP-Oracle family and exact solvers on the fly;
- obstacles on two terrains at 0.5M faces, with disks covering 5, 10 and 20% of the area and six walls, with errors reported by detour ratio.

[This section will report these results.]

7 Discussion

The experiments support the main claim: a terrain distance oracle can answer in a fraction of a microsecond and certify every answer, with errors far below those of the Steiner-graph oracles and an index one to two orders of magnitude smaller. The certificate is what makes the index usable as a database operator. A kNN or range query can decide most candidates from the intervals alone, and a user who needs a guaranteed answer can ask for UB, which is the length of an actual path, or for an exact refinement only when the interval is too wide.

7.1 Limitations

Asymptotic size. The uncapped index stores $\Theta(n^{1.5})$ labels, as the top cuts have $\Theta(\sqrt{n})$ portals and every vertex stores them. The capped variant is near-linear, but its $(1+\varepsilon)$ guarantee relies on the exact refine for the rare close pairs across a capped cut, and Djidjev and Sommer's oracle is asymptotically smaller in theory.

Build. The build is dominated by one exact sweep per portal. It is parallel, but uncapped its cost grows like $n^{1.5}$, and it is the most expensive part of the index at millions of faces.

Slopes and weights. λ enters the near radius, the window and the ellipse. Steep terrains do not change λ, but heavily weighted terrains with a small $w_{\min}$ widen all three, and on weighted terrains the labels are Steiner distances, so we certify only the upper bound.

Static terrains. The index is static. Terrains that change, the setting of UP-Oracle, would need the affected cuts rebuilt.

8 Conclusion

We introduced the Certified Cut Index, a distance oracle for terrain surfaces that stores exact geodesic distances to portals sampled on the cut lines of a kd hierarchy and returns every distance with a certified interval within a factor $1 \pm \varepsilon$. The window lemma lets the cuts be sampled, the ellipse lemma makes the labels exact and cheap to compute, and the 1-Lipschitz label rows give the lower bound. The index answers in a fraction of a microsecond, is far more accurate and smaller than the existing terrain oracles, and extends to arbitrary points, proximity queries, weighted terrains and obstacles. Future work includes the large-scale evaluation, updates for changing terrains, and closing the gap to near-linear space without the refine step.

AI use statement

We used generative AI tools to help write and test parts of the implementation and the experiment scripts, to summarise related

literature, and to draft and edit parts of this manuscript. The author reviewed the code, checked the cited works against the original sources and verified the proofs and the experimental results, and takes full responsibility for the content.

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A Proofs

We use the notation of Section 4. In the implementation a portal that falls within 10^{-7} of a face edge is moved along the line by about 10^{-5} edge lengths before it is inserted, so a run end may not be a portal. It then leaves an end stretch [run end, p_k] of arc e_k, and the lower bound also takes the term $f_k - 2e_k$.

Lemma A.1 (crossing). *Every curve on T between two points on opposite sides of ℓ_C meets ℓ_C.*

Proof. The a-coordinate of its chart projection is continuous, below v at one end and above at the other. □

Proof of Lemma 4.3 with end stretches. If the crossing x^* lies in an end stretch with portal p_k, then $f_k \le d(s,x^*)+2A(x^*,p_k)+d(x^*,t) \le d(s,t)+2e_k$, so $f_k - 2e_k \le d(s,t)$. The interval case is as in the main text. □

Proof of Theorem 4.4. Let $|s-t| > R$ and $d = d(s,t)$, so $d \ge \lambda|s-t| > g/\varepsilon$ by (1). Let x^* be the crossing of a shortest path in the interval $[p_j, p_{j+1}]$ with arc gap A_j, and $a = A(p_j, x^*)$.

Upper bound. $\min(f_j, f_{j+1}) \le d + \min(2a, 2(A_j - a)) \le d + A_j$. If the interval lies in I_C then $A_j \le g < \varepsilon d$. Otherwise every point of the interval, x^* included, has offset at least c_j from I_C, so $d \ge 2\lambda c_j$ by Lemma 4.1, and (2) gives $A_j \le \max(g, 2\varepsilon\lambda c_j) \le \varepsilon d$. In an end stretch, $f_k \le d + 2e_k$, and $2e_k$ is far below $g < \varepsilon d$.

Lower bound. Every $f \ge d$, so an interval term satisfies $\frac{1}{2}(f_j + f_{j+1}) - A_j \ge d - A_j$, which is at least $(1-\varepsilon)d$ if $A_j \le \varepsilon d$. Otherwise $A_j > \varepsilon d > g$, so the interval is a tail interval with $A_j \le 2\varepsilon\lambda c_j$, hence $c_j > d/(2\lambda)$. Both of its portals p have offset at least c_j from B_M, so by (1) $f(p) \ge \lambda(|s-p| + |p-t|) \ge 2\lambda c_j$, and the term is at least $2\lambda c_j(1-\varepsilon) > (1-\varepsilon)d$. An end-stretch term is $f_k - 2e_k \ge d - 2e_k > (1-\varepsilon)d$. The chord term is a lower bound, so $\mathrm{LB} \ge (1-\varepsilon)d$. □

Proof of Theorem 4.5. Lemma 4.3 holds for any portal set, so $\mathrm{LB} \le d \le \mathrm{UB}$. The proof of Theorem 4.4 used $g < \varepsilon d$ only through the gap inside I_C; with gap g_C it gives $\mathrm{UB} \le (1+\varepsilon)d$ whenever $g_C \le \varepsilon d$, and the query refines every pair for which this is not certified, that is every pair with $\mathrm{UB} > (1+\varepsilon)\mathrm{LB}$. A refined pair is computed by one exact sweep on the faces meeting $\mathrm{bbox}(s,t)$ dilated by $\mathrm{UB}/(2\lambda)$, which contains every path of length at most UB by Lemma 4.2, so it is exact.

For the size, a cut has at most k portals in its box. In a tail the gap is g_C until $2\varepsilon\lambda c \ge g_C$, which takes $O(g_C/(2\varepsilon\lambda)/g_C) = O(1/\varepsilon)$ portals, and from then on the offset grows by a factor at least $1 + 2\varepsilon\lambda/\sigma$ per portal, up to the window length μ_C/λ. With $\mu_C \approx D_C$ and $g_C \approx \sigma D_C/k$ for a cut of length D_C, this takes $O\big(\frac{\sigma}{\varepsilon\lambda}\log\frac{2\varepsilon k}{\sigma}\big) =$

$O(\varepsilon^{-1} \log k)$ portals. A vertex lies in $O(\log n)$ cuts, which gives the label count. □

Proof of Theorem 4.6. $\mathrm{UB}(p, q)$ is the cost of a path from p to a corner, then a path certified by $\mathrm{UB}(a_i, b_j)$, then to q. For the lower bound, $d(a_i, b_j) \le d(a_i, p) + d(p, q) + d(q, b_j) \le w_f |P - A_i| + d(p, q) + w_{f'} |Q - B_j|$, so every term of the maximum is at most $d(p, q)$. Pairs whose interval is wider than $1 + 2\varepsilon$, and near pairs, are computed exactly on an ellipse sub-mesh with both points inserted as vertices. □